Mateusz Płoszaj-Mazurek

Architect, PhD · Building researcher and computational designer

Warsaw, Poland · remote across time zones · mateuszploszajmazurek@gmail.com github.com/Curiosit

Architect with ten years of practice in Denmark and Poland, a PhD in digital design methods and two years building design software for architects. I research daylight and building regulation at Warsaw University of Technology and build computational tools that make building performance part of design, not a check at the end.

Experience

2017–now
Researcher and lecturer
Faculty of Architecture, Warsaw University of Technology · doctoral candidate from 2017, academic staff since 2021

- Daylight regulation, simulation and surrogate-model research; see publications.
- Teaches Grasshopper and multi-criteria building optimization (400+ students so far); runs a second-year design studio.

2024–2026
Software engineer, interim product owner
One Click LCA, Helsinki · remote

- Built features for Carbon Designer 3D, an early-design tool for architects (TypeScript, React, Three.js, AG Grid).
- Interim product owner in the second half of 2025; took the Free Drawing feature from specification to release.
- Wrote methodology specifications.

2024
Contract developer
enxess, Liechtenstein

Web viewer for IFC models of historical building objects (Three.js, React, TypeScript).

2023–2024
Founder
SLAD.AI

Self-funded venture building early-design carbon estimation software; designed, built and tested with users.

2014–2024
Architect, Certified Passive House Designer
Bjerg Arkitektur · Hjørring 2014–16, Warsaw 2016–24

- Single- and multi-family housing, schools, kindergartens and offices, from 100 m^2 to 15,000 m^2.
- All design phases: concept, permit and construction documentation, and architectural supervision on site.
- Code compliance in practice under Danish and Polish regulations and Passive House certification; daylight simulation studies with VELUX.
- 2022–2024: architecture, sustainability and software consultant, building in-house design tools; co-author of the Polish whole-life carbon roadmap.

Selected projects

2025–26
OmegaFlux
Per-point daylight prediction from geometric view descriptors with gradient-boosted trees, exported to ONNX and run in the browser. Dataset generation, held-out validation against Radiance, transfer to unseen room shapes. curiosit.github.io/OmegaFlux

2026
Solaie: sunlight-access checker for Revit
own product, in testing

Revit 2027 add-in (C#, .NET, WPF) with a compiled Python analysis engine: model scanning, geometry export, vectorised ray tracing against context meshes, signed time-limited licences and an HTML report for building authorities. solaie.pl

2025
Free Drawing, Carbon Designer 3D
One Click LCA

Led a feature letting architects sketch building massing directly in the browser, from user problem through specification to release.

2021
Whole Life Carbon Roadmap for Poland
PLGBC, WorldGBC, EBRD

Co-author, with Alicja Kuczera, of the national roadmap for decarbonising the built environment by 2050.

Publications

A workflow for LLM-assisted IFC data enrichment to support circular reuse of large-panel building systems in Poland

with A. M. Tofiluk · Journal of Building Engineering 120, 115397 (2026) · doi · open access

Artificial Intelligence and Digital Tools for Assisting Low-Carbon Architectural Design: Merging the Use of Machine Learning, Large Language Models, and Building Information Modeling for Life Cycle Assessment Tool Development
with E. Ryńska · Energies 17(12), 2997 (2024) · doi

Methods to Optimize Carbon Footprint of Buildings in Regenerative Architectural Design with the Use of Machine Learning, Convolutional Neural Network, and Parametric Design
with E. Ryńska and M. Grochulska-Salak · Energies 13(20), 5289 (2020) · doi **136 citations**

Analytic radiometric descriptors enable geometry-flexible per-sensor daylight prediction without learned spatial encodings preprint
SSRN (2026) · code github.com/Curiosit/OmegaFlux · software doi:10.5281/zenodo.21781259

Akademik na Solcu, Warsaw
Project publication, with M. Czechowicz · Architektura-Murator 4/2025, pp. 54–64

Przedszkole pasywne, Reguły
Passive House kindergarten, project publication, with D. Sibińska, J. Sokołowski et al. · Architektura-Murator 5/2023, pp. 20–34

Skills

Methods	Evolutionary and multi-objective optimization, surrogate modelling, gradient-boosted trees, dataset and validation design, parametric experiments
Building domain	Climate-based daylight (sDA, ASE, UDI, EN 17037), insolation, Polish and Danish building regulations, Passive House, building carbon footprint
Design tools	Rhino and Grasshopper, Rhino.Inside.Revit, Revit, ArchiCAD, Ladybug and Honeybee, Radiance, THERM, IFC, That Open Company tools
Code	Python (NumPy, pandas, scikit-learn, XGBoost, ONNX), C#, TypeScript, React, Three.js, Git, CI/CD; daily AI-assisted development with Claude Code, including six months on a production codebase
Languages	Polish (native), English (fluent), Danish (communicative), Chinese (basic)

Education

2022 — **PhD in Architecture**
Warsaw University of Technology
Digital methods of supporting the architectural design process and the analysis of the carbon footprint of buildings: parametric optimization and machine-learning methods for design.

2014–2016 — **MSc in Architecture**
Aalborg University, Denmark

2023–2024 — **Certifications**
Master BIM Software Developer (That Open Company, 2024) · Harvard CS50x (2023) · Certified Passive House Designer

Grants and awards

2024–25 — **Principal investigator, IDUB research grant** · Circular economy in architecture: reuse of prefabricated elements and digital design techniques · Warsaw University of Technology, CPR-IDUB/238/Z01/2024

2024 — **First prize, Best Mashup Project** · AEC Hackathon, Copenhagen

2023 — **Competition win: energy-efficient multi-storey housing in Poland** · Winning entry by Studio Widoki, with Bjerg Arkitektur as consultant · Carbon footprint calculations by Bjerg Arkitektur: prefabricated timber, about 6 kg CO_2e/m^2 per year over 50 years

2022 — **PLGBC Green Building Awards: best green building without certification** · Passive House kindergarten in Podkowa Leśna, Bjerg Arkitektur; award received on behalf of the practice. Also a distinction for an educational building.

2022 — **Distinction for the doctoral thesis** · Innowator Mazowsza competition

Teaching

7 years — **Introduction to Grasshopper**
"Cyfrowe techniki geometryczne": parametric modelling and multi-criteria optimization of energy, daylight and carbon.

2020– — **iBIM postgraduate programme (Interdisciplinary BIM Studies)**
Introductory and advanced Grasshopper, environmental and daylight analysis with Rhino.Inside.Revit and Grasshopper, Python and C# scripting, for practising architects and engineers. Hybrid.

Talks and panels

2023 — **Pro-environmental design: training sessions for architects** · Saint-Gobain · two sessions, Jura Krakowsko-Częstochowska and Żnin, October 2023